DAAN: A Deep Autoencoder-Based Augmented Network for Blind Multilinear Hyperspectral Unmixing

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***Abstract*— In recent years, deep learning (DL) has accelerated the development of hyperspectral image (HSI) processing, expanding the range of applications further. As a typical model of unsupervised DL, the autoencoder framework has been extensively applied for spectral unmixing due to its strong representation ability and scalability. Nowadays, most DL-based unmixing approaches adopt the linear mixture model (LMM) to estimate pure spectral signatures (endmembers) and their corresponding abundance fractions. However, since sunlight scattering is an inevitable physical phenomenon, the spectral mixture problem is inherently nonlinear. Moreover, most existing nonlinear unmixing approaches focus exclusively on spectral information, neglecting the spatial distribution of materials and the intrinsic correlation between pixels, making it challenging to explore latent features. To address these issues, this article develops a new deep autoencoder-based augmented network (DAAN). The proposed DAAN employs the multilinear mixture model (MLMM) to handle the nonlinear influence caused by multiple scattering. Meanwhile, the proposed DAAN constraints homogenous smoothing in the autoencoder architecture, enabling the aggregation of intrinsic correlations by means of spatial relationships to enhance the performance of abundance estimation. We achieve unsupervised nonlinear hyperspectral unmixing by combining spectral and spatial information. The effectiveness and advantages of DAAN are confirmed by several experiments with synthetic and real HSI datasets. The results indicate that the proposed method outperforms other DL-based unmixing approaches. The source codes of the proposed DAAN will be provided in the following link https://github.com/yuanchaosu/TGRS-daan.**

***Index Terms*— Autoencoder, deep learning (DL), hyperspectral remote sensing, nonlinear unmixing.**

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I. Introduction

HYPERSPECTRAL remote sensing has been widely applied in many fields, such as precision agriculture, forestry, mineral exploration, environmental testing, and military applications [1], [2], [3], [4]. Hyperspectral imagery sensors capture detailed spectral information and intricate spatial characteristics of target scenes. This enables them to effectively discriminate between different land-cover classes, thereby advancing the field of quantitative remote sensing [5], [6], [7]. However, the spatial resolution of hyperspectral images (HSIs) has been a critical impediment in some applications [8], [9], [10]. HSIs usually contain many mixed pixels due to their limited spatial resolution [9]. Although the spatial resolution of HSIs has been significantly improved with the advancement of imaging spectrometer technology, the mixed pixel problem does not completely disappear with the increase in spatial resolution [11], [12], [13]. Therefore, processing mixed pixels is significant for developing hyperspectral remote sensing applications [14], [15].

A. Classic Unmixing

Hyperspectral unmixing is an effective analytical technique for handling mixed pixels. The primary objective of unmixing is to obtain the pure spectral signatures (endmembers) contained in each pixel and their associated fractional abundances [16], [17], [18]. Spectral mixture models are the basis for implementing unmixing. Nowadays, the spectral mixture models can be divided into two main categories: linear spectral mixture model (LSMM) and nonlinear spectral mixture model (NSMM) [19], [20], [21], [22].

The LSMM assumes that the spectral signatures of mixed pixels are linear combinations of endmember signatures [23], [24]. Considering the LSMM's simplicity, most existing unmixing approaches were developed under the LSMM [25]. Nevertheless, the LSMM operates under an ideal condition that the photon reaching the sensor interacts with just one

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material in a macroscopic scale [26]. Since the LSMM is a rough assumption and does not hold in practice, it may not meet the needs of certain applications [27], [28], [29].

NSMMs contain various models that delineate the interactions of solar radiation when scattered by diverse materials [30]. These models intricately characterize the interactions at both intimate or multilayered levels, thereby offering some in-depth explorations to consider the scattering phenomena [31], [32]. No matter if the interactions are modeled under intimate or multilayered levels, the related spectral mixture models belong to NSMMs, and several works have been developed for implementing nonlinear unmixing [33], [34], [35].

The intimate mixture assumptions hold that photons are emitted by molecules from one substance and then absorbed by molecules of a different substance, potentially leading to the emission of additional photons [36]. Typically, the assumptions considering intimate mixtures are more faithful to reflect the imaging mechanism of the instrument [37]. Nevertheless, it should be noted that these models are highly sensitive to certain intrinsic parameters, often determined by the sensor's geometric positioning [38]. This reliance on external parameters poses significant challenges in executing the inversion process [39]. Furthermore, the characterization of intimate mixtures also increases the complexity of the model dramatically [40].

In the case of multilayered mixtures, the corresponding models consider that interactions among scatterers occur across various layers, resulting in an infinite sequence of powers of products of reflectance [41]. The multilayered mixture assumptions can be formalized into a combination of the linear mixture model (LMM) and additional scattering terms, where the scattering terms represent the interactions of photons from different materials [42], [43]. Regrettably, most multilayered mixture assumptions require detailed knowledge of the physical interactions corresponding to multiple materials, limiting their application in practice [44], [45]. Recently, bilinear mixture models (BMMs) have been developed to deal with the complexities of multilayered mixtures, aiming to ensure generalization, such as the Fan model [46], the generalized bilinear model [47], and the hierarchical BMM [16]. However, BMMs only reach the modeling of the second-order scattering of photons. Since higher-order scattering is a natural phenomenon, BMMs ignore some interactions in the real world. The multilinear mixture model (MLMM) extends BMMs to include all orders of interactions, which simulates the scattering interactions in the imaging process more accurately than BMMs [48]. Although the algorithm in [48] can drive the MLMM to achieve nonlinear unmixing, it is a supervised method that needs endmembers to be provided by users. Furthermore, the algorithm in [48] does not consider the spatial information during the implementation of unmixing.

B. DL-Based Unmixing

In recent years, deep learning (DL) has significantly impacted the development of HSI processing, and unmixing is no exception [49], [50]. Nowadays, DL-based unmixing methods have gradually become the mainstream for handling mixed pixels in the hyperspectral remote sensing field [51], [52], [53]. As a DL model, autoencoder is currently the most used architecture for designing unmixing methods [54], [55]. An autoencoder consists of an encoder and a decoder, and its hidden layers can obtain high-level features from observed data, making it particularly well-suited for hyperspectral unmixing because abundance estimation involves finding a low-dimensional representation [56]. However, most autoencoder-based unmixing approaches only focused on spectral information while neglecting spatial relationships of pixels [55].

Recently, convolutional neural networks (CNNs) have provided some schemes for collaboratively utilizing spatial and spectral information of HSIs, depending on the advantage of acquiring local correlations between pixels [57]. The advantage of CNNs has fostered many new unmixing methods, such as the CNN autoencoder unmixing (CNNAEU) [58], minimum simplex convolutional network (MiSiCNet) [15], deep interpretable fully CNN (DIFCNN) [59], Bayesian fully CNN (BCUN) [60], and 3DCNN autoencoder network [61]. In the aforementioned methods, the CNNAEU, MiSiCNet, DIFCNN, and BCUN rely on the LSMM and cannot eliminate scattering interactions when implementing unmixing. Moreover, most CNN-based methods only employ 2-D convolution kernels to consider spatial information from HSIs, which cannot explore the latent correlations from spatial and spectral information collaboratively [61]. Although the 3DCNN autoencoder network adopts the 3-D convolution kernel to consider spatial–spectral structures of HSIs, the method does not consider the problem caused by multiple scattering.

C. Motivation

It is a difficult problem to realize unsupervised removal of the impact of multiple scattering when implementing nonlinear unmixing. Although the MLMM provides a path to process multiple scattering, there is no effective DL framework to drive it for blind unmixing. Moreover, combining spatial and spectral information to improve the performance of nonlinear unmixing is significant for developing elaborate and intelligent subpixel interpretation technologies.

D. Overview and Contributions

This article develops a new deep autoencoder-based augmented network (DAAN) to realize blind nonlinear unmixing under the MLMM, which considers multiple scattering and combines spatial and spectral information in the unmixing. Specifically, the network framework contains two encoders (Encoder 1 and Encoder 2) and a decoder. Encoder 1 combines 3-D convolutions with a Gaussian constraint to concern the estimations of endmembers and abundances, while Encoder 2 concentrates on predicting the impact of multiple scattering. Note that Encoder 2 needs to be driven by Encoder 1 during the training. The Gaussian constraint can assist convolutions in further acquiring local correlations between homogeneous pixels, enhancing the smoothness of abundances. The decoder is used for obtaining endmembers and abundances by blind signal separation (BSS). In the proposed DAAN, a Softmax

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is adopted to ensure the abundance nonnegativity constraint (ANC) and the abundance sum-to-one constraint (ASC). Moreover, we add $L_{1/2}$ regularization to suppress the interferential features from noise, further enhancing the robustness of the deep network.

The main contributions of our work can be summarized as follows.

1) The proposed DAAN achieves blind nonlinear unmixing by considering multiple scattering.
2) This work provides an unsupervised DL framework for using the MLMM.
3) A new technique for combining spatial and spectral information is designed to assist nonlinear unmixing.

The remainder of this article is organized as follows. Section II briefly reviews two related works: LSMM and MLMM. Section III describes the proposed DAAN in detail. In Section IV, we use several HSIs (including synthetic and real datasets) to evaluate the effectiveness of our DAAN as compared to other state-of-the-art (SOTA) approaches. Section V concludes the article with some remarks and hints at plausible future research lines.

II. Related Works

The MLMM is an extension of the LSMM that handles the nonlinear effects caused by multiple scattering. The proposed DAAN establishes two encoders to divide the MLMM into its linear and multilinear components when implementing blind unmixing. Therefore, the related spectral mixture models involve the LSMM and MLMM.

A. Linear Spectral Mixture Model

The LSMM has gained widespread application in spectral unmixing, primarily due to its simplicity and broad applicability [51]. The model assumes that the mixed pixels are linear combinations of endmembers, and it has two physical constraints (ANC and ASC). Its mathematical formulation can be written as

$$\begin{aligned} &\mathbf{Y} = \mathbf{E}\mathbf{A} + \Upsilon \\ &\text{s.t.}\, \mathbf{A} \geq 0,\ \mathbf{1}_m^\top \mathbf{A} = \mathbf{1}_n^\top \end{aligned} \tag{1}$$

where $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_n] \in \mathbb{R}^{c\times n}$ is an HSI with n pixels and c bands. In (1), $\mathbf{E} = [\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_m] \in \mathbb{R}^{c\times m}$ represents the endmember matrix with m pure materials, and $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n] \in \mathbb{R}^{m\times n}$ is the abundance matrix, and Υ denotes the noise matrix. Endmember signatures are also nonnegative because their shapes depend on the reflectivity or radiance of the materials [40]. When using the LSMM for unmixing, the loss function is generally defined as

$$(\mathbf{E}, \mathbf{A}) = \arg\min_{\mathbf{A}} \|\mathbf{Y} - \mathbf{E}\mathbf{A}\|_F^2 \tag{2}$$

where $\|\cdot\|_F^2$ denotes the Frobenius norm.

B. Multilinear Mixture Model

The MLMM is illuminated by BMMs and extends the scope of interactions to all orders, modeling multiple scattering in imaging [48]. This model is based on the discrete Markov chain of the reflection process and assumes incoming light rays from the source will interact with at least one material. Afterward, in each interaction with an endmember, the photon has a transition probability p for further interactions and a probability $(1-p)$ of escaping the scenario and being received by the sensors. The probability of interacting with the mth endmember is proportional to its abundance fraction. The light intensity will change according to that material's albedo when scattering. When neglecting the minor differences between the albedo and the endmember spectrum, the albedo is approximated to be equal to the endmember spectrum [48]. The process is illustrated in Fig. 1, where we use two objects in the diagram. According to [45] and [48], the MLMM can be formulated as follows:

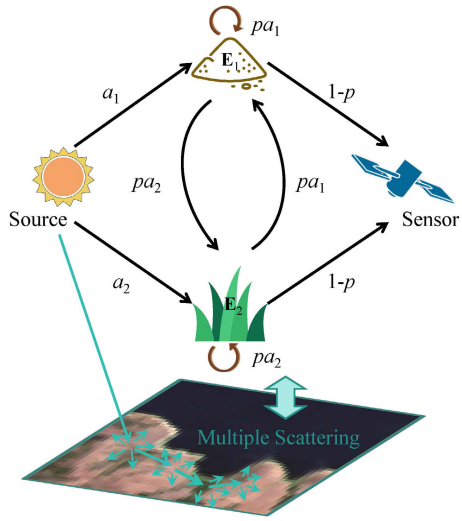


Fig. 1. Illustrating the transition probabilities between source, endmembers, and sensor. The example uses two objects: $\mathbf{E}_1$ and $\mathbf{E}_2$.

$$\begin{aligned} \mathbf{y} = &(1-p)\sum_{k=1}^{m} \mathbf{e}_k a_k + (1-p)p\sum_{k=1}^{m}\sum_{j=1}^{m}(\mathbf{e}_k \odot \mathbf{e}_j)a_k a_j \\ &+ (1-p)p^2\sum_{k=1}^{m}\sum_{j=1}^{m}\sum_{i=1}^{m}(\mathbf{e}_k \odot \mathbf{e}_j \odot \mathbf{e}_i)a_k a_j a_i + \ldots \end{aligned} \tag{3}$$

where p is regarded as the scattering parameter, and $\odot$ represents the Hadamard product. Following the derivations in [48], when neglecting noise, (3) can be simplified as

$$\mathbf{y} = \frac{(1-p)\sum_{k=1}^{m} a_k \mathbf{e}_k}{1 - p\sum_{k=1}^{m} a_k \mathbf{e}_k} \tag{4}$$

where a_k represents the proportion of the endmember $\mathbf{e}_k$ in the pixel. It can be observed that the MLMM will degrade to the LSMM without considering noise if $p = 0$.

III. Proposed Method

In the DAAN framework, we establish two encoders to concern linear mixtures and multiple scattering, respectively. Encoder 1 is designed to consider the linear component and adopts 3-D convolutional layers to capture the local correlations by the spatial–spectral structure of HSI. Meanwhile, we employ a Gaussian constraint in Encoder 1 to ensure the

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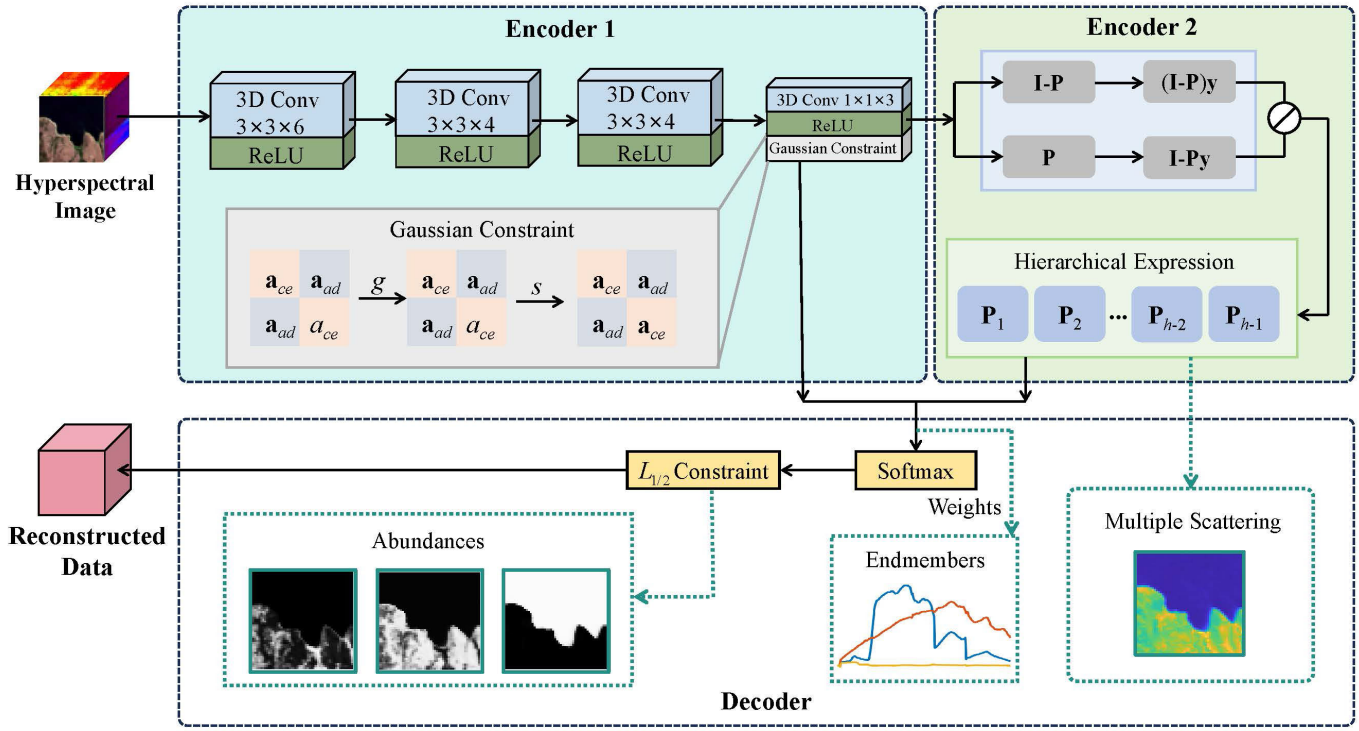


Fig. 2. Flowchart of the proposed DAAN, containing two encoders and a decoder. Encoder 1 concerns the linear component of the MLMM, while Encoder 2 is designed to handle the impact of multiple scattering. The symbol ⊘ represents the division operator. The decoder is established to implement BSS to complete unmixing, obtaining endmembers and abundances after removing scattering interactions.

smoothing of estimated abundances associated with homogeneous pixels. Encoder 2 is established to estimate the impact of multiple scattering by the hierarchical expression. In the decoder of DAAN, we employ a Softmax layer to ensure ANC and ASC. Moreover, we adopt the $L_{1/2}$ regularization to enhance noise resistance and ensure robustness when meeting low-quality scenarios. The network parameters are updated by backward gradient propagation during learning. Finally, the obtained endmembers and abundances are the unmixing results after removing the impact of scattering interactions. Fig. 2 illustrates the flowchart of the proposed DAAN.

A. Encoder 1 for Linear Mixture Component

Encoder 1 stacks three 3-D convolutional layers. The first layer adopts a $3 \times 3 \times 6$ convolutional kernel. The second and third layers use the same $3 \times 3 \times 4$ convolutional kernel, while the fourth layer adopts a $1 \times 1 \times 3$ convolutional kernel. The first three 3-D convolutional layers are employed to explore local correlations from spatial–spectral information in an HSI, while the last convolutional layer is utilized to reduce information redundantly and reshape the size of abundances. Moreover, we adopt the ReLU as the activation to avoid negative values in training.

Moreover, we adopt the Gaussian constraint to ensure the smoothness of abundances, considering that homogeneous pixels tend to be distributed in adjacent regions in an HSI. Let $\mathbf{g}^{(r)}$ be the Gaussian weights between the central pixel and the adjacent pixels, which can be expressed as

$$\mathbf{g} \Leftarrow g_r = e^{-\frac{\left\|\mathbf{a}_{\text{ce}} - \mathbf{a}_{\text{ad}}^{(r)}\right\|^2}{2\sigma^2}} \quad \forall \left\{\mathbf{a}_{\text{ce}}, \mathbf{a}_{\text{ad}}^{(r)}\right\} \in \mathbf{K}^{(4)} \tag{5}$$

where $r = 1, 2, \ldots, q$ is the number of adjacent pixels, $\mathbf{a}_{\text{ce}}$ is the central pixel, $\mathbf{a}_{\text{ad}}^{(r)}$ is the rth adjacent pixel, and $\mathbf{K}^{(4)} \in \mathbb{R}^{c \times n}$ denotes the spatial–spectral features by the fourth 3-D convolutional layer. We assume that $\mathbf{d}$ is the weight of adjacent pixels, and it can be written as

$$\mathbf{d} = \sum_{r=0}^{q} \mathbf{g}^{(r)} \mathbf{a}_{\text{ad}}^{(r)}. \tag{6}$$

Let $s^{(t)}$ denote the difference weight between the central and the adjacent pixels, which can be expressed as

$$s^{(t)} = \left\|\mathbf{a}_{\text{ce}}^{(t)} - \mathbf{d}^{(t)}\right\|_F^2 \tag{7}$$

where $\mathbf{a}_{\text{ce}}^{(t)}$ is the abundance of the tth central pixel, $\mathbf{d}^{(t)}$ is the tth weighted adjacent pixels, $t = 1, 2, \ldots, n$, and $\|\cdot\|_F^2$ represents the Frobenius norm. The difference weight $\mathbf{s}^{(t)}$ will be updated with the implementation of DAAN. A diagonal matrix $\mathbf{S} \in \mathbb{R}^{n \times n}$ is composed of $s^{(t)}$, which can enhance the smoothness of abundances, and the regularization can be written as

$$\mathbf{B} = \mathbf{K}^{(4)}\mathbf{S} \tag{8}$$

where $\mathbf{B} \in \mathbb{R}^{m \times n}$ is a latent feature matrix, and $\mathbf{K}^{(4)}$ is the feature matrix output from the fourth 3-D convolutional layer. The matrix $\mathbf{S}$ is then multiplied by the feature matrix output from the fourth 3-D convolutional layer to obtain smooth latent features for the subsequent process. Fig. 3 illustrates the procedure that ensures the smoothness of abundances by the Gaussian constraint.

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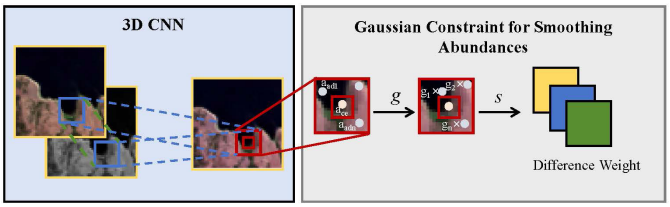


Fig. 3. Ensuring the smoothness of abundances by the Gaussian constraint. A sliding window is designed to capture the local information between the central pixel's abundance $\mathbf{a}_{ce}$ and the adjacent pixels' abundances $\mathbf{a}_{ad}$. Gaussian weights can be obtained by (5). The difference weight between the central and adjacent pixels will be calculated by (7) and updated with raining Encoder 1.

B. Encoder 2 for Multilinear Interaction Component

Encoder 2 hierarchically implements the MLMM to handle elaborate scattering interactions. According to (4), the matrix expression of the MLMM can be written as

$$\mathbf{Y} = \left((\mathbf{I}-\mathbf{P})(\mathbf{EA})^{\top}\right)^{\top} / \left(\Phi - \left(\mathbf{P}(\mathbf{EA})^{\top}\right)^{\top}\right) \tag{9}$$

where $\mathbf{I} \in \mathbb{R}^{n\times n}$ is an identity matrix, $\mathbf{P} \in \mathbb{R}^{n\times n}$ represents a diagonal matrix that contains n scattering parameters, and $\Phi \in \mathbb{R}^{c\times n}$ represents a matrix of all ones. Since $\mathbf{P}$ is a diagonal matrix, (9) can be rewritten as

$$\begin{aligned}\mathbf{Y} &= \mathbf{EA}(\mathbf{I}-\mathbf{P})/(\Phi - \mathbf{EAP}) \\ &= \frac{\mathbf{EA}-\mathbf{EAP}}{\Phi - \mathbf{EAP}}.\end{aligned} \tag{10}$$

Following the strategy in [62], the hierarchical expression of (10) can be written as

$$\mathbf{Y} = \frac{\mathbf{EA}_1\mathbf{A}_2,\ldots,\mathbf{A}_h - \mathbf{EA}_1\mathbf{A}_2,\ldots,\mathbf{A}_h\mathbf{P}_h}{\Phi - \mathbf{EA}_1\mathbf{A}_2,\ldots,\mathbf{A}_h\mathbf{P}_h} \tag{11}$$

where $\mathbf{A}_1\mathbf{A}_2,\ldots,\mathbf{A}_h$ are hierarchical expressions of abundances, $\mathbf{A}_1 \in \mathbb{R}^{m\times n}$, and $\mathbf{A}_2,\mathbf{A}_3,\ldots,\mathbf{A}_h \in \mathbb{R}^{n\times n}$. In Encoder 2, scattering parameters can be hierarchically estimated by hidden layers, and the procedure can be approximated as

$$\begin{aligned}&\mathbf{P}_{h-1} \approx \mathbf{A}_h\mathbf{P}_h \\ &\vdots \\ &\mathbf{P}_2 \approx \mathbf{A}_3,\ldots,\mathbf{A}_h\mathbf{P}_h \\ &\mathbf{P}_1 \approx \mathbf{A}_2,\ldots,\mathbf{A}_h\mathbf{P}_h\end{aligned} \tag{12}$$

where $\mathbf{P}_1\mathbf{P}_2,\ldots,\mathbf{P}_h$ are the hidden layers in the block. The abundance and the scattering parameter matrices can be obtained by the following scheme:

$$\begin{aligned}&\mathbf{A} = \mathbf{A}_1\mathbf{A}_2,\ldots,\mathbf{A}_h \\ &\mathbf{P} = \mathbf{P}_h.\end{aligned} \tag{13}$$

C. Loss Function

The proposed DAAN considers the smoothness of abundances and the impact of multiple scattering. Therefore, the optimization can be formalized as

$$\begin{aligned}\psi_{\mathbf{A},\mathbf{P}} = \alpha_1 &\left(\arg\min_{\mathbf{A}} \frac{1}{2}\left(\left\|\mathbf{Y} - \frac{\mathbf{EA}_1\mathbf{A}_2,\ldots,\mathbf{A}_h - \mathbf{EA}_1\mathbf{A}_2,\ldots,\mathbf{A}_h\mathbf{P}_h}{\Phi - \mathbf{EA}_1\mathbf{A}_2,\ldots,\mathbf{A}_h\mathbf{P}_h}\right\|\right)\right) \\ &+ \lambda f(\mathbf{A}) + \alpha_2\left(\arg\min_{\mathbf{P}} \frac{1}{2}\left(\left\|\mathbf{Y} - \frac{\mathbf{EA} - \mathbf{EAP}_h}{\Phi - \mathbf{EAP}_h}\right\|\right)\right)\end{aligned} \tag{14}$$

where α_1, λ, and α_2 are regularization coefficients, and $f(\mathbf{A})$ is the $L_{1/2}$ regularization. In (14), α_1 and α_2 are associated with two sub-optimizations concerning the linear mixture component and multilinear interaction component, respectively. The $L_{1/2}$ regularization is adopted to enhance the performance of abundance estimation

$$f(\mathbf{A}) = \left(\sum_{k=1}^{m}\sum_{u=1}^{n}|a_{ku}|^{1/2}\right)^2 \tag{15}$$

where a_{ku} is an abundance fraction. We adopt the spectral angle distance (SAD) as a complement to measure information loss in the training

$$\Gamma_{\text{SAD}} = \frac{1}{n}\sum_{u=1}^{n}\arccos\left(\frac{\mathbf{y}_u\hat{\mathbf{y}}_u}{\|\mathbf{y}_u\|_2\|\hat{\mathbf{y}}_u\|_2}\right) \tag{16}$$

where $\hat{\mathbf{y}}_u$ is the reconstruction of the uth pixel $\mathbf{y}_u$. The loss function of DAAN is expressed as

$$\Gamma = \psi_{\mathbf{A},\mathbf{P}} + \Gamma_{\text{SAD}}. \tag{17}$$

D. Decoder for Obtaining Results

In the decoder, we employ the Softmax function to ensure ANC and ASC

$$\mathbf{A}_1 \Leftarrow \text{Softmax}(\mathbf{B}) \tag{18}$$

where $\mathbf{B}$ is obtained from the fourth convolutional layer. The decoder adopts Adam [63] as the optimizer to implement gradient descent to update endmembers, hierarchical expressions, and scattering parameters. Let $\mathbf{G}_{(\varphi)}$ be the gradient of $\mathbf{E}$. Following [63], the biased first and second moments in φth iteration can be updated as

$$\begin{cases}\Theta_{(\varphi)} = \beta_1\Theta_{(\varphi-1)} + (1-\beta_1)\mathbf{G}_{(\varphi)} \\ \Psi_{(\varphi)} = \beta_2\Psi_{(\varphi-1)} + (1-\beta_2)(\mathbf{G}_{(\varphi)} \odot \mathbf{G}_{(\varphi)})\end{cases} \tag{19}$$

where $\Theta_{(\varphi)}$ and $\Psi_{(\varphi)}$ are the biased first and second moments, respectively. In (19), β_1 and β_2 are the exponential decay rates. The bias corrections can be expressed as

$$\begin{cases}\hat{\Theta}_{(\varphi)} = \dfrac{\Theta_{(\varphi)}}{\mathbf{1}-\beta_1^{\varphi}} \\ \hat{\Psi}_{(\varphi)} = \dfrac{\Psi_{(\varphi)}}{\mathbf{1}-\beta_2^{\varphi}}\end{cases} \tag{20}$$

where β_1^{φ} and β_2^{φ} denote β_1 and β_1 to the power φ. According to [63], the update of $\mathbf{E}$ can be expressed as

$$\mathbf{E}_{(\varphi+1)} \Longleftarrow \mathbf{E}_{(\varphi)} - \frac{\eta}{\sqrt{\hat{\Psi}_{(\varphi)}} + \Omega} \odot \hat{\Theta}_{(\varphi)} \tag{21}$$

where η is the learning rate, and Ω consists of small constants to avoid division by zero. Similar to (19)–(21), we can update $\mathbf{A}_l$ and $\mathbf{P}$, therefore, the optimization can be expressed simply as

$$\mathbf{E}, \mathbf{A}_l, \mathbf{P} \Longleftarrow \text{Adam}(\mathbf{E}, \mathbf{A}_l, \mathbf{P}) \tag{22}$$

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Algorithm 1 Pseudocode of the Proposed DAAN

Input: HSI $\mathbf{Y}$.
Output: Endmembers $\mathbf{E}$, Abundances $\mathbf{A}$, Scattering Parameters $\mathbf{P}$.

```
Initialize E by a pure-pixel assumption algorithm;
Implement the 3-D convolutional layers to aggregate spatial-spectral correlations;
for Epochs do
    repeat
        Obtain Gaussian weight by (5);
        Update the difference weight between the central and adjacent pixels by (7);
        Acquire the latent features by (8);
        Calculate the abundance matrix by (13);
    until Reach convergence in (16);
    Acquire A_1 preliminarily by (18);
    Initialize A_2, A_h and P_h randomly;
    repeat
        Update E, A_l, P by (22);
        Normalize P by (23);
        Calculate the abundances by (13).
    until Reach convergence in (17);
end
until Stop conditions
```

where $\mathbf{A}_l \in \mathbb{R}^{n\times n}$, $l = 2, 3, \ldots, h$. Note that the settings of β_1, β_2, η, and Ω follow the recommendations in [63]. Moreover, we use normalization to avoid outliers when estimating $\mathbf{P}$, and the process can be written as

$$p = \frac{p - p_{\min}}{p_{\max} - p_{\min}} \tag{23}$$

where $p_{\min}$ and $p_{\max}$ are the minimum and maximum in $\mathbf{P}$. Afterward, we can obtain abundance matrix $\mathbf{A}$ by (13). Finally, the proposed DAAN can output endmembers, abundances, and scattering parameters. The pseudocode of DAAN is given in Algorithm 1.

The proposed DAAN is a model-driven unmixing method that is based on the MLMM and employs a hierarchical scheme to solve this physical model. Moreover, as an unsupervised nonlinear unmixing approach, DAAN does not require prior knowledge, leading to better universality and convenience than most data-driven and supervised methods.

IV. Experimental Results and Analysis

In this section, several experiments with synthetic and real HSIs evaluate the effectiveness of our DAAN. For quantitative analysis, we use the SAD to assess extracted endmembers, and the related mathematical expression is

$$\mathrm{SAD}\left(\mathbf{e}_i, \widehat{\mathbf{e}}_i\right) = \arccos\left(\frac{\left[\mathbf{e}_i, \widehat{\mathbf{e}}_i\right]}{\|\mathbf{e}_i\| \cdot \left\|\widehat{\mathbf{e}}_i\right\|}\right) \tag{24}$$

where $\mathbf{e}_i$ is the ith estimated endmember by DAAN, and $\widehat{\mathbf{e}}_i$ is an ideal spectrum in a library. Moreover, we utilize the root-mean-square error (RMSE) to evaluate abundances as follows:

$$\mathrm{RMSE}\left(\mathbf{a}_u, \widehat{\mathbf{a}}_u\right) = \frac{1}{n}\sum_{u=1}^{n}\sqrt{\left\|\mathbf{a}_u - \widehat{\mathbf{a}}_u\right\|_2} \tag{25}$$

where $\mathbf{a}_u$ is the estimated abundance by DAAN, and $\widehat{\mathbf{a}}_u$ is the ground-truth abundance. The reconstruction error (RE) is regarded as a complement that can reflect the loss of information after implementing the proposed approach

$$\mathrm{RE}\left(\{\mathbf{y}_u\}_{u=1}^{n}, \{\widehat{\mathbf{y}}_u\}_{u=1}^{n}\right) = \frac{1}{n}\sum_{u=1}^{n}\sqrt{\left\|\mathbf{y}_u - \widehat{\mathbf{y}}_u\right\|_2} \tag{26}$$

where $\mathbf{y}_u$ is the observed data, and $\widehat{\mathbf{y}}_u$ represents the related reconstruction.

In order to evaluate the competitiveness of DAAN, we use some SOTA methods as baselines, including vertex component analysis (VCA) [64], fully constrained least squares (FCLSs) [65], unmixing using deep image prior (UnDIP) [66], 3-D CNN (3DCNN) [67], robust nonnegative matrix factorization (rNMF) [68], deep autoencoders with multitask learning for bilinear unmixing (DMBU) [16], and MLMM [48].

1) VCA is based on the pure-pixel assumption and is designed for endmember extraction [64]. In this article, we use VCA to initialize endmembers for DAAN.
2) FCLS is a classical algorithm for calculating abundances after obtaining endmembers, and it considers ANC and ASC [65].
3) UnDIP combines deep image prior with CNN to solve a regularized inverse problem for improving unmixing performances [66].
4) 3DCNN is the abbreviation of 3-D CNN convolutional network in this article. The method can acquire spatial–spectral priors to enhance unmixing performances under an autoencoder framework [67].
5) rNMF regards the impact of the scattering phenomenon as an outlier and does not consider specific NSMM [68].
6) DMBU employs two deep autoencoders with multitask learning to achieve nonlinear unmixing under the hierarchical BMM [16].
7) MLMM is a conventional algorithm that can solve the MLMM to achieve unmixing [48]. However, the algorithm is supervised instead of blind unmixing, because it does not extract endmembers.

Note that VCA, FCLS, UnDIP, and 3DCNN belong to linear unmixing approaches. Conversely, the aforementioned rNMF, DMBU, and MLMM serve for nonlinear unmixing. For the baselines, we use the default settings of hyperparameters published in their codes.

In this work, the proposed DAAN is implemented in PyTorch. The recommended experimental environment is Python version 3.8.10 and PyTorch version 2.0.1. All experiments are conducted on a Microsoft Windows server with a GPU (NVIDIA GeForce RTX 3080 Ti).

A. Experiments With Synthetic Data

To evaluate the effectiveness of DAAN, we constructed synthetic data that uses three spectra of minerals from the

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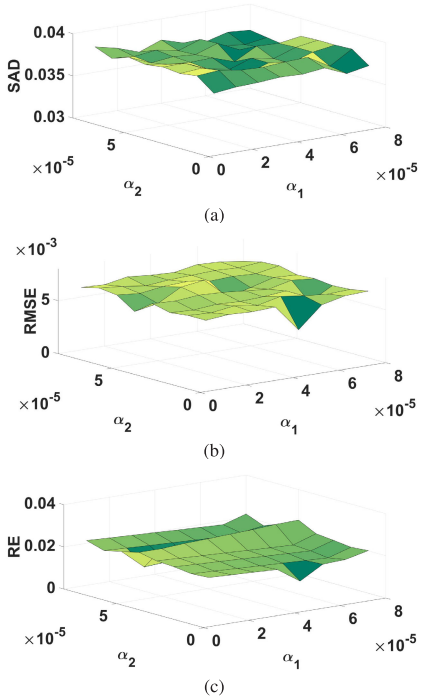


Fig. 4. Results of the proposed DAAN with respect to hyperparameters α_1 and α_1. (a) SAD. (b) RMSE. (c) Mean RE.

USGS spectral library[1] as endmembers. We set fractional matrices as abundances and use the MLMM to generate the synthetic data. The synthetic HSI has 676 pixels with 176 bands. Moreover, we can adjust the signal-to-noise ratio (SNR) to test the robustness of DAAN. In our experiments with the synthetic data, the SNRs are set to 20, 30, and 50 dB.

1) Hyperparameter Analysis: In this test, we evaluate the effects by setting different hyperparameters. Experimental results related to the hyperparameters are displayed in Fig. 4. It is observed that Fig. 4 provides the differences in results due to setting different hyperparameters. In Fig. 4, we can find that the proposed DAAN can reach the best performance when the hyperparameters are set to $\alpha_1 = 1 \times 10^{-5}$ and $\alpha_2 = 5 \times 10^{-5}$. The hyperparameter $\lambda = 1 \times 10^{-2}$ needs to be set to a value relatively larger than α_1 and α_2 because it is associated with the smoothness of abundances. In this experiments, we adopt $\alpha_1 = 1 \times 10^{-5}$, $\alpha_2 = 5 \times 10^{-5}$, and $\lambda = 1 \times 10^{-2}$.

Considering the hierarchical expressions in (11)–(13), we evaluate the impact of the hierarchical expressions, and the results are listed in Table I, where we set SNR = 50 dB for this synthetic dataset. It can be seen that the results corresponding to $h = 4$ achieve the best unmixing performances in Table I. Moreover, as we increase the value of h, the computation burden of DAAN also increases. Nevertheless, the unmixing performances are extremely similar if $h > 2$. After comprehensively considering efficiency and performance, we set $h = 3$ for subsequent experiments.

2) Ablation Experiment: The ablation experiment assesses all terms in the loss function, corresponding to (14), (16), and (17), respectively. The experiments were conducted on a dataset with SNR = 50 dB. The results of the ablation experiments are given in Table II. The results indicate that these terms play positive roles in unmixing. As an illustration, Fig. 5 presents the changes of the RE along with iterations. Fig. 5 further confirms the contributions of the terms by visualizing the changes of the RE under different conditions.

3) Convolutional Kernel Settings: Considering that convolutional operations are indispensable in DAAN, this experiment evaluates the sizes of convolutional kernels to influence unmixing performance. Table III lists the results of DAAN by setting different convolutional sizes. For each group test, from top to bottom, the kernel sizes correspond to the first, second, third, and fourth convolutional layers, respectively. It can be observed that the results in the fourth group are better than others. For convenience, subsequent experiments also adopt the settings from the fourth group in Table III.

4) Comparisons: In order to demonstrate the competitiveness of DAAN, we use the synthetic dataset to test the aforementioned baselines to acquire their unmixing results. Fig. 6 shows the abundance maps estimated by different methods. It can be observed from Fig. 6 that our proposed DAAN method provides abundance maps that closely resemble the ground truth. Furthermore, Table IV lists the quantitative analysis results of different unmixing methods when using the synthetic dataset. In Table IV, the results of RE, RMSE, and SAD are obtained after ten independent Monte Carlo runs. In Table IV, we can see that the proposed DAAN can obtain

[1]https://speclab.cr.usgs.gov/spectrallib.html

TABLE I

MEAN SADS (IN RADIANS), RMSES, RES, AND COMPUTATION TIME (IN SECONDS) OBTAINED BY DAAN UNDER SETTING DIFFERENT LAYERS OF THE HIERARCHICAL EXPRESSION

	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 5$
Mean SAD	0.0381	0.0374	0.0365	0.0363	0.0368
RMSE	0.0078	0.0053	0.0046	0.0042	0.0049
RE	0.0209	0.0117	0.0113	0.0110	0.0115
Time	14.22	15.17	16.02	18.52	20.73

TABLE II

SAD, RMSE, AND RE RESULTS OF USING DIFFERENT LOSS FUNCTIONS IN THE ABLATION EXPERIMENT

Loss Functions	Mean SAD	RSME	RE
$\psi(\mathbf{A}, \mathbf{P})$	0.0416	0.0047	0.0115
Γ_{SAD}	0.0443	0.0058	0.0117
$\psi(\mathbf{A}, \mathbf{P}) + \Gamma_{SAD}$	0.0362	0.0037	0.0110

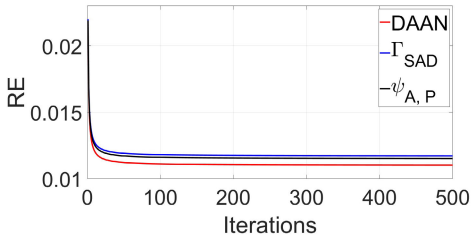
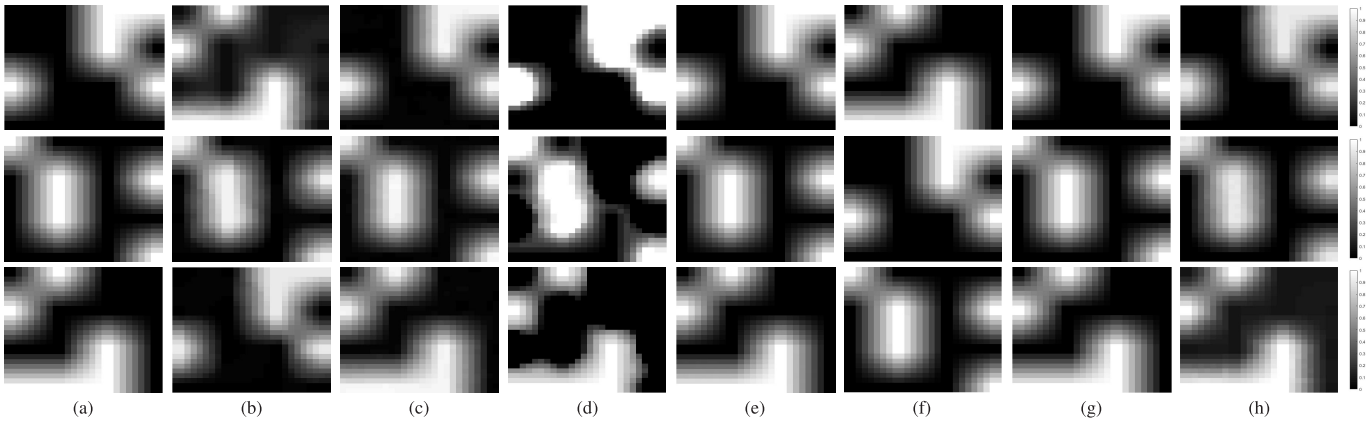


Fig. 5. Changes of the RE along with iterations. Red line: DAAN; blue line: Γ_{SAD}; black line: $\psi(\mathbf{A}, \mathbf{P})$.

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(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 6. Abundance maps obtained by different methods on the synthetic data. (Top) Endmember #1. (Middle) Endmember #2. (Bottom) Endmember #3. (a) Ground truth. (b) VCA+FCLS. (c) UnDIP. (d) 3DCNN. (e) rNMF. (f) DMBU. (g) MLMM. (h) DAAN.

TABLE III

THIS EXPERIMENT CONDUCTS A DETAILED ANALYSIS OF CONVOLUTIONAL KERNELS IN FOUR LAYERS WHEN PROCESSING 50 dB SYNTHETIC DATA. IT PRESENTS THE NUMERICAL VALUES OF AVERAGE SAD (IN RADIANS), RE, AND RMSE. EACH GROUP USES DIFFERENT KERNEL SIZES FOR EACH LAYER

Scenario	Kernel Size	Mean SAD	RMSE	RE
1^{st}Group	$3\times3\times3$ $3\times3\times3$ $3\times3\times3$ $1\times1\times1$	0.0393	0.0047	0.0137
2^{nd}Group	$3\times3\times4$ $3\times3\times4$ $3\times3\times4$ $1\times1\times3$	0.0389	0.0041	0.0129
3^{rd}Group	$3\times3\times6$ $3\times3\times6$ $3\times3\times6$ $1\times1\times3$	0.0385	0.0043	0.0132
4^{th}Group	$3\times3\times6$ $3\times3\times4$ $3\times3\times4$ $1\times1\times3$	0.0362	0.0037	0.0110

better unmixing performances than other methods. However, in Table IV, we can observe that the proposed DAAN has no advantage in terms of computation time compared with traditional unmixing methods. Fig. 7 compares estimated endmembers by DAAN with the ground truth. In Fig. 7, we can see that the estimated endmembers are extremely similar to library spectra.

B. Experiments With Real HSIs

In this article, we use four real HSI datasets to confirm the effectiveness and competitiveness of DAAN, including the Moffett Field, Samson, Jasper Ridge, and Shuangfeng datasets. In the experiment with the Moffett Field data, the proposed DAAN adopts $\alpha_1 = 2\times10^{-5}$, $\alpha_2 = 3\times10^{-4}$, and $\lambda = 1\times10^{-3}$. In the test with the Samson data, we set $\alpha_1 = 1\times10^{-6}$, $\alpha_2 = 2\times10^{-5}$, and $\lambda = 3\times10^{-3}$ in DAAN. In the experiment with the Jasper Ridge data, we set $\alpha_1 = 4\times10^{-6}$, $\alpha_2 = 3\times10^{-4}$, and $\lambda = 2\times10^{-4}$. For the test with the Shuangfeng data, DAAN uses $\alpha_1 = 5\times10^{-7}$, $\alpha_2 = 4\times10^{-6}$, and $\lambda = 1\times10^{-5}$.

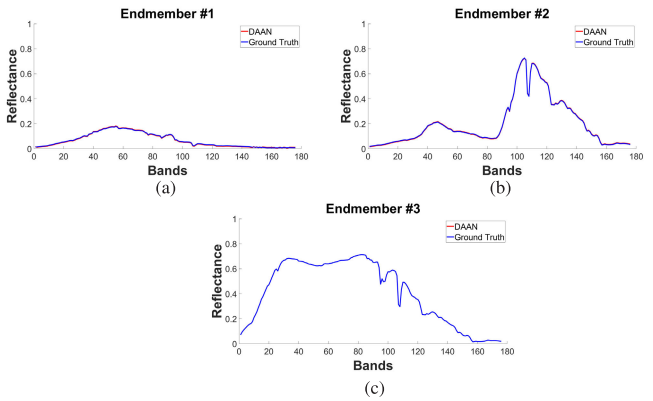


Fig. 7. Comparison of extracted endmembers by the proposed DAAN and the ground truth on the synthetic dataset. Blue lines: ground truth. Red lines: estimated endmembers. (a) Endmember #1. (b) Endmember #2. (c) Endmember #3.

1) Experiment With the Moffett Field Data: The Moffett Field data is the most commonly used dataset for testing nonlinear unmixing. The dataset is collected by the AVIRIS sensor in the Moffett area at the southern end of San Francisco, California, USA [69]. This dataset covers a wavelength range from 0.4 to 2.5 μm, and each pixel contains 224 bands. The dataset contains 50 $\times$ 50 pixels, containing three objects: soil, vegetation, and water. After removing noisy bands, the experiment used the remaining 189 bands. The selected region comprises three endmembers that are characteristic of the coastal area in the image: vegetation, water, and soil. This subscale is widely utilized in remote sensing to evaluate nonlinear unmixing methods and has wide applications.

The abundances acquired from different methods are displayed in Fig. 8. It is observed from Fig. 8 that the results of our DAAN are better than others. Considering that there are no attached ground truths with the dataset, the experiment only uses the RE as the metric to evaluate the accuracy of the obtained unmixing results. The quantitative comparisons are listed in Table V. From Table V, we can observe that the proposed DAAN obtained the best RE value. Fig. 9 shows the endmembers estimated by the proposed DAAN.

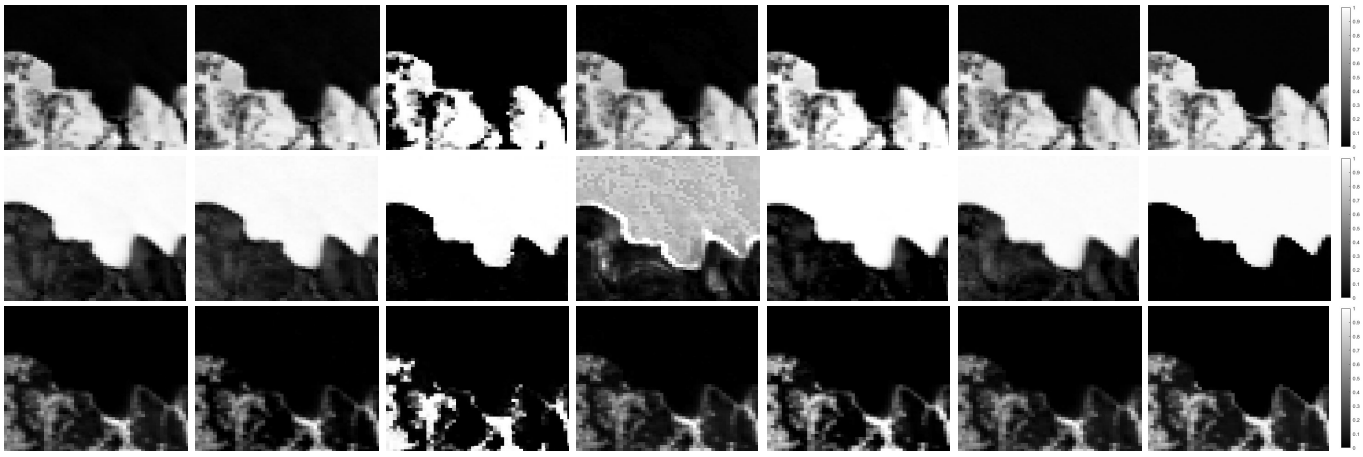
In this article, we utilize $\mathbf{Y} - \mathbf{YP}$ and reshape it as a cube to visualize the multiple scattering distribution in an

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TABLE IV

MEAN SADS (IN RADIANS), RES, RMSES, AND COMPUTATIONAL TIME (SECONDS) ALONG WITH THE STANDARD DEVIATIONS OBTAINED FROM TEN INDEPENDENT MONTE CARLO RUNS FOR THE CONSIDERED SYNTHETIC DATA. THE BEST RESULTS ARE DISPLAYED IN BOLD

Type		LSMM-based			NSMM-based			
Method		VCA+FCLS	UnDIP	3DCNN	rNMF	DMBU	MLMM	DAAN
SNR=20 dB	Mean SAD	0.1798±0.42%	0.1701±0.32%	0.1559±0.38%	0.1664±0.70%	0.1661±0.50%	0.1744±0.35%	**0.1541**±0.13%
	RMSE	0.0501±0.15%	0.0439±0.13%	0.0421±0.12%	0.0439±0.14%	0.0435±0.07%	0.0452±0.13%	**0.0401**±0.15%
	RE	0.0812±0.44%	0.0801±0.56%	0.0752±0.42%	0.0812±0.41%	0.0734±0.47%	0.0702±0.45%	**0.0642**±0.35%
	Time	**6.13**	46.87	22.21	20.05	16.20	16.52	20.80
SNR=30 dB	Mean SAD	1571±0.45%	0.1568±0.48%	0.1556±0.65%	0.1601±0.65%	0.1560±0.62%	0.1565±0.42%	**0.1507**±0.31%
	RMSE	0.0092±0.12%	0.0086±0.15%	0.0081±0.17%	0.0089±0.32%	0.0087±0.23%	0.0087±0.29%	**0.0072**±0.15%
	RE	0.0503±0.46%	0.0478±0.32%	0.0467±0.53%	0.0477±0.32%	0.0470±0.62%	0.0471±0.32%	**0.0452**±0.16%
	Time	**5.58**	47.12	22.44	19.97	16.14	15.97	21.02
SNR=50 dB	Mean SAD	0.0399±0.52%	0.0382±0.46%	0.0370±0.45%	0.0382±0.47%	0.0378±0.32%	0.0375±0.23%	**0.0362**±0.21%
	RMSE	0.0063±0.16%	0.0053±0.15%	0.0051±0.33%	0.0061±0.33%	0.0047±0.23%	0.0050±0.23%	**0.0037**±0.11%
	RE	0.0230±0.19%	0.0200±0.22%	0.0170±0.20%	0.0188±0.45%	0.0190±0.09%	0.0177±0.32%	**0.0110**±0.14%
	Time	**5.36**	45.19	21.94	19.33	15.69	16.15	20.71



(a) (b) (c) (d) (e) (f) (g)

Fig. 8. Abundance maps obtained by different methods on the Moffett Field data. (Top) Soil. (Middle) Vegetation. (Bottom) Water. (a) VCA+FCLS. (b) UnDIP. (c) 3DCNN. (d) rNMF. (e) DMBU.(f) MLMM. (g) DAAN.

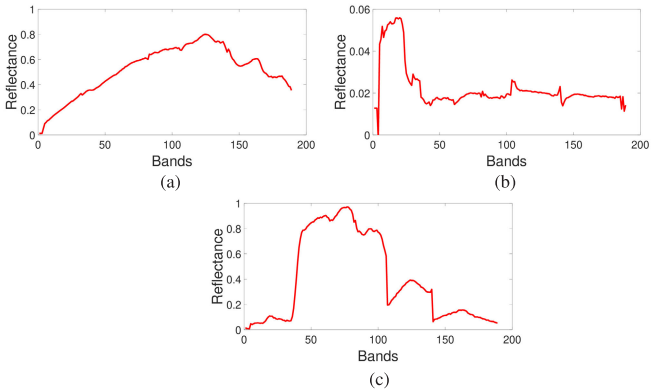


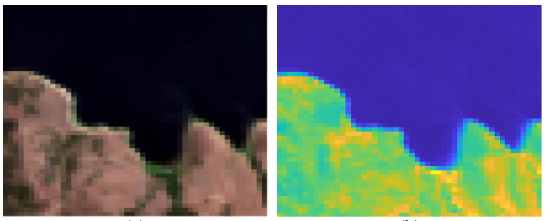
Fig. 9. Endmembers estimated by the proposed DAAN in the Moffett Field data. (a) Soil. (b) Water. (c) Vegetation.

TABLE V

RES AND COMPUTATIONAL TIME (SECONDS) OBTAINED BY DIFFERENT METHODS ON THE MOFFETT FIELD DATA. THE BEST RESULTS ARE DISPLAYED IN BOLD

Type	LSMM-based			NSMM-based			
Method	VCA+FCLS	UnDIP	3DCNN	rNMF	DMBU	MLMM	DAAN
RE	0.0206	0.0202	0.0190	0.0199	0.0187	0.0192	**0.0181**
Time	**20.13**	89.01	67.42	52.14	39.24	49.32	43.10

image. Fig. 10 displays the scattering interaction distribution on Moffett Field data estimated by DAAN. To the best of



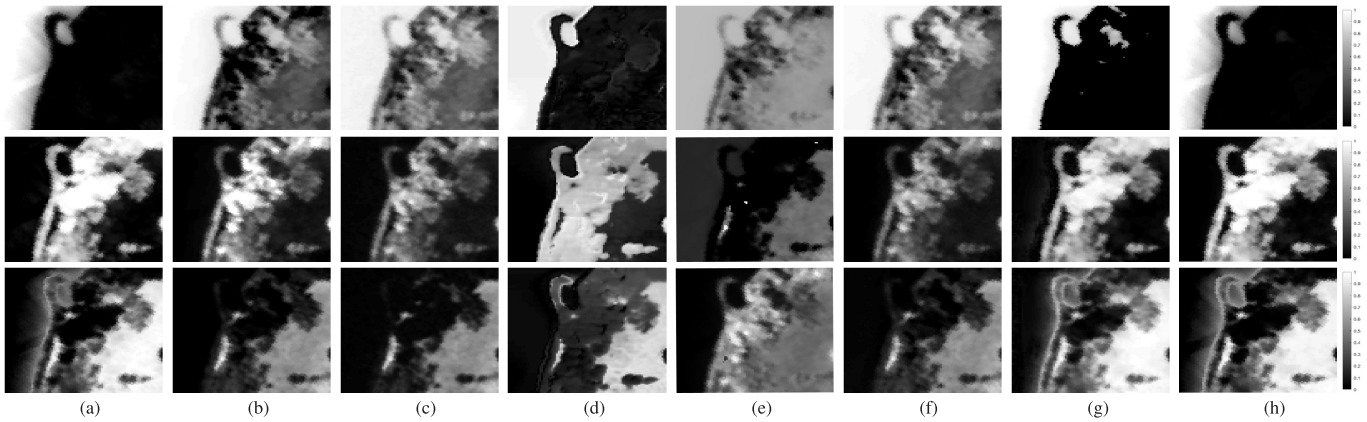
(a) (b)

Fig. 10. Interaction distributions estimated by DAAN on the Moffett Field data. Bright yellow means a large value, while dark blue is a small value. (a) RGB composition. (b) Estimation of DANN.

our knowledge, the multiple scattering phenomenon usually occurs in areas where the material is highly mixed. We can see that the estimation in Fig. 10 is in accordance with geoscience commonsense compared with the RGB map of the Moffett Field data. The results in Figs. 8 and 10, and Table V prove the effectiveness and competitiveness of DAAN. Meanwhile, it can be observed that the proposed DAAN does not increase the obvious computational time compared with other DL-based/nonlinear unmixing methods.

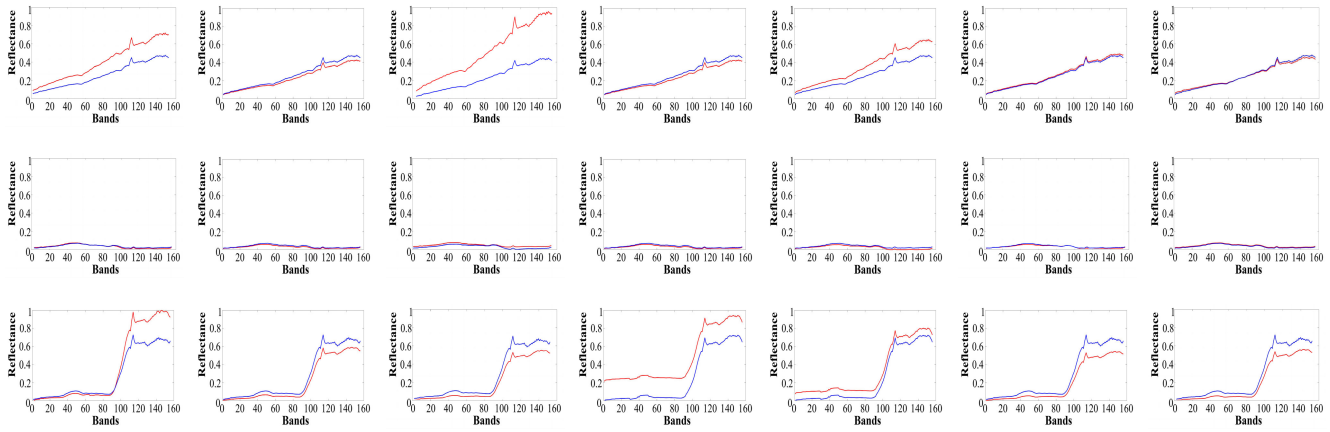
2) Experiment With the Samson Data: The Samson dataset was acquired using the SAMSON sensor and has 95× 95 pixels with 156 bands, and the spectral range is from 0.401 to 0.889 μm [70]. In this scenario, there are three objects: soil,

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(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 11. Abundance maps obtained by different unmixing algorithms on the Samson data. (Top) Water. (Middle) Tree. (Bottom) Soil. (a) Ground reference. (b) VCA+FCLS. (c) UnDIP. (d) 3DCNN. (e) rNMF. (f) DMBU. (g) MLM. (h) DAAN.



(a) (b) (c) (d) (e) (f) (g)

Fig. 12. Comparison of extracted endmembers by different unmixing methods on the Samson data. Blue lines: ground reference. Red lines: estimated endmembers. (Top) Soil. (Middle) Water. (Bottom) Tree. (a) VCA. (b) UnDIP. (c) 3DCNN. (d) rNMF (e) DMBU. (f) MLMM. (g) DAAN.

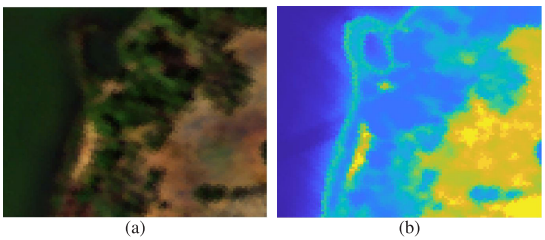
TABLE VI

SADs (in Radians) RMSEs, and REs Obtained by Different Methods on the Samson Data, Where the Best Unmixing Results Are Displayed in Bold

Method		VCA+FCLS	UnDIP	3DCNN	rNMF	DMBU	MLMM	DAAN
SAD	Soil	0.0910	0.0108	0.1133	0.0108	0.0709	0.0108	**0.0103**
	Water	0.0122	0.0109	0.0106	0.0107	0.0105	0.0106	**0.0104**
	Tree	0.1127	0.0516	0.0512	0.1410	0.0503	0.0178	**0.0153**
Mean SAD		0.0720	0.0244	0.0584	0.0542	0.0439	0.0131	**0.0120**
RMSE		0.1421	0.1301	0.1317	0.1523	0.1305	0.1299	**0.1256**
RE		0.1003	0.0802	0.0823	0.0959	0.0734	0.0645	**0.0510**

tree, and water. Moreover, the Samson data attaches the ground reference that can be used to assess the accuracy of unmixing quantitatively, including endmembers and abundances.

Fig. 11 shows the abundance maps estimated by different unmixing methods on the Samson data. We can observe that the abundance maps estimated by DAAN are closer to the ground truth than other unmixing methods. Fig. 12 shows the endmember spectral estimated by different unmixing methods. It is seen that the endmembers extracted by DAAN are better than others. The comparisons in Figs. 11 and 12 can visually verify the effectiveness of DAAN. Table VI lists the quantitative comparison of results obtained by different methods. In Table VI, the SAD, RMSE, and RE corresponding to DAAN are best in all results, further highlighting the proposed approach's competitiveness. Additionally, Fig. 13



(a) (b)

Fig. 13. Interaction distributions estimated by DAAN on the Samson data. Bright yellow means a large value, while dark blue is a small value. (a) RGB composition. (b) DANN.

illustrates the distribution of scattering interactions estimated by DAAN on the Samson image. As it is well-known, scattering interactions mainly occur where different materials are mixed. In the scenario of the Samson data, the scattering distribution estimated by DAAN meets the expectation because there are no strong effects on the water.

3) Experiment With the Jasper Ridge Data: The Jasper Ridge data was collected by the AVIRIS sensor in central California, United States [70]. The dataset comprises 100 × 100 pixels, and the scenario contains four materials: water, soil, tree, and road. The original data has 224 bands, spanning a spectral range from 0.38 to 2.5 μm. In this experiment, we retained 198 bands because some bands were removed to

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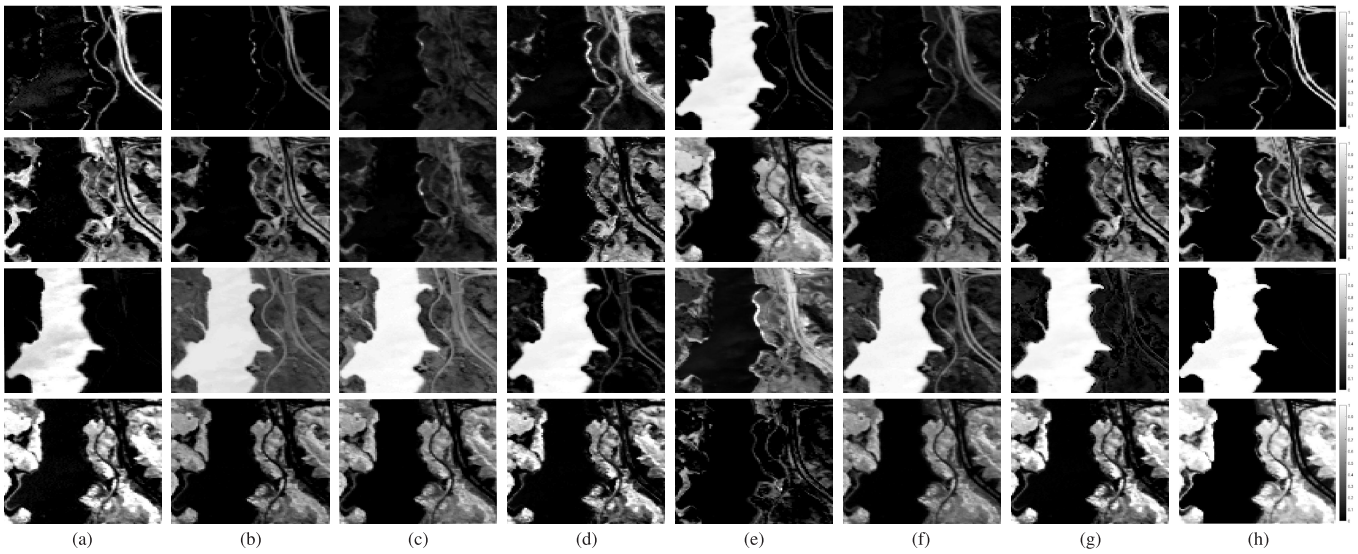


Fig. 14. Abundance maps obtained by different unmixing algorithms on the Jasper Ridge data. (Top) Water. (Middle) Tree. (Bottom) Soil. (a) Ground reference. (b) VCA+FCLS. (c) UnDIP. (d) 3DCNN. (e) rNMF. (f) DMBU. (g) MLM. (h) DAAN.

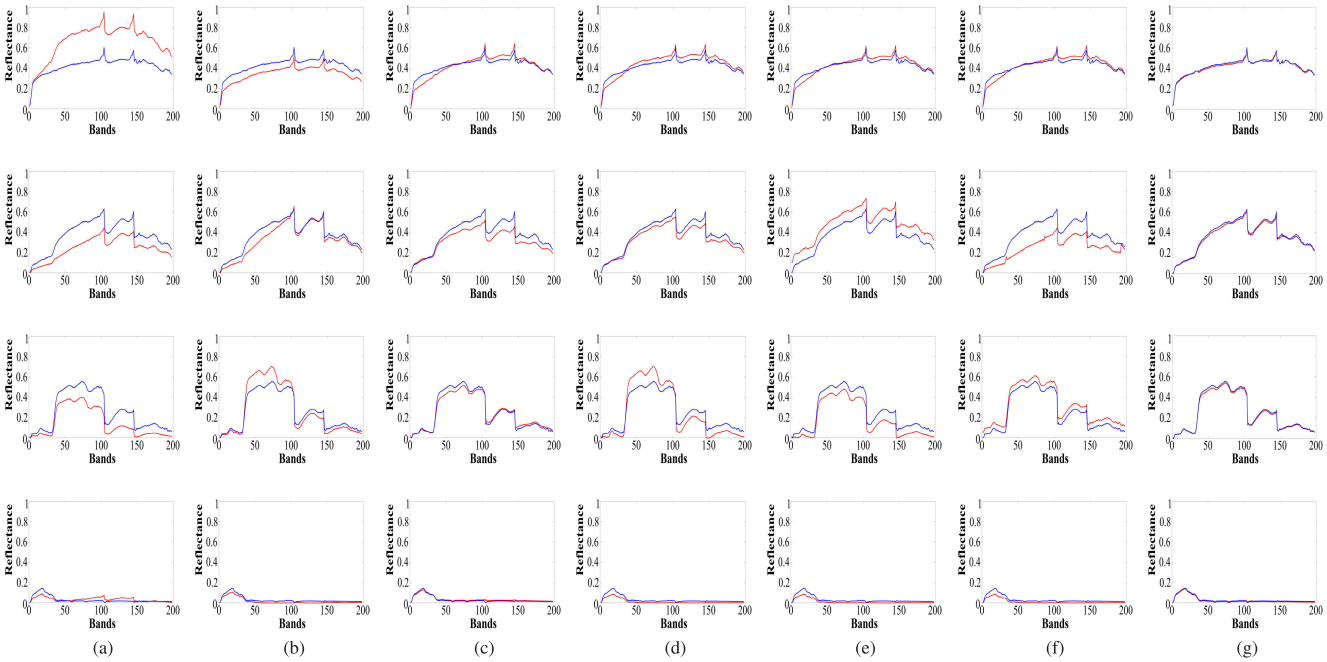


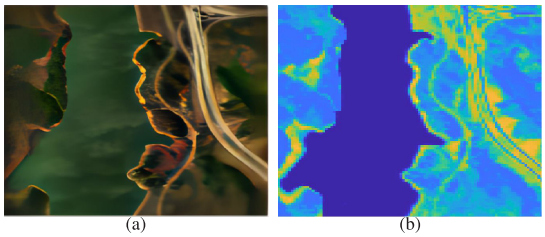
Fig. 15. Comparison of extracted endmembers by different unmixing methods on the Jasper Ridge data. Blue lines: ground reference. Red lines: estimated endmembers. (Top) Soil. (Middle) Water. (Bottom) Tree. (a) VCA. (b) UnDIP. (c) 3DCNN. (d) rNMF (e) DMBU. (f) MLMM. (g) DAAN.

reduce the effects of significant water vapor and atmospheric interferences. Moreover, the Jasper Ridge data also attaches its ground reference.

Fig. 14 provides the abundance map obtained by unmixing approaches. It can be seen that the abundances obtained by DAAN are better than other methods' results compared with the ground truth. The endmember signatures extracted by different unmixing methods are shown in Fig. 15. The comparisons of spectra in Fig. 15 strongly confirm the effectiveness of DAAN because its estimated endmembers are closer to the ground truth. Table VII provides the unmixing quantitative comparisons, including SADs, RMSEs, and REs acquired by different methods. In Table VII, the results of DAAN are better than those achieved by other unmixing methods. Fig. 16 displays the distribution of scattering effects estimated by DAAN. The results in Figs. 14–16, and Table VII further confirm the effectiveness and competitiveness of the proposed DAAN.

4) Experiment With the Shuangfeng Data: The Shuangfeng data covers a part of Shuangfeng County, Hunan Province, China. The dataset was acquired by the ZY-1 02D satellite equipped with a visible shortwave infrared hyperspectral camera (AHSI). The dataset has 166 bands, and the spectral range is from 0.40 to 2.50 μm. The spectra contain 90 shortwave infrared bands and 76 near-infrared bands. The scenario has 100 $\times$ 100 pixels and surveys three materials: water, vegetation, and soil. The data has undergone preprocessing, including atmospheric correction and other steps. However,

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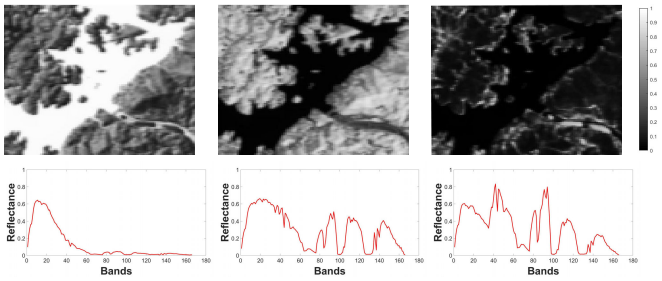
(a) (b)

Fig. 16. Interaction distributions estimated by DAAN on the Jasper Ridge data. Bright yellow means a large value, while dark blue is a small value. (a) RGB composition. (b) DANN.

TABLE VII

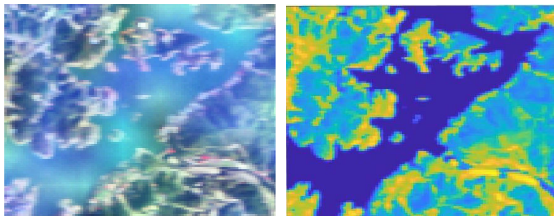
SADs (IN RADIANS) RMSEs, AND REs OBTAINED BY DIFFERENT METHODS IN THE JASPER RIDGE DATA, WHERE THE BEST UNMIXING RESULTS ARE DISPLAYED IN BOLD

Method		VCA+FCLS	UnDIP	3DCNN	rNMF	DMBU	MLMM	DAAN
SAD	Road	0.2031	0.0865	0.0769	0.0734	0.0672	0.0621	**0.0576**
	Soil	0.1432	0.0861	0.0879	0.0723	0.0876	0.0845	**0.0675**
	Tree	0.0914	0.0907	0.0762	0.0904	0.0792	0.0783	**0.0621**
	Water	0.0828	0.0734	0.0763	0.0712	0.0771	0.0823	**0.0701**
Mean SAD		0.1301	0.0842	0.0793	0.0768	0.0778	0.0768	**0.0643**
RMSE		0.1920	0.1502	0.1465	0.1803	0.1494	0.1488	**0.1362**
RE		0.1583	0.1202	0.0702	0.1202	0.1105	0.0965	**0.0610**



(a) (b) (c)

Fig. 17. Abundance maps and endmembers estimated by the proposed DAAN on the Shuangfeng data. (a) Water. (b) Vegetation. (c) Soil.



(a) (b)

Fig. 18. Interaction distributions estimated by DAAN on the Shuangfeng data. Bright yellow means a large value, while dark blue is a small value. (a) RGB composition. (b) DANN.

the Shuangfeng data has no reference signatures and abundances as ground truth. Therefore, the experiment with the Mediterranean data can only provide a qualitative analysis.

Fig. 17 illustrates the RGB composition and the range of multiple scattering distributions estimated by DAAN. We can observe that the estimated interactions by DAAN mainly occur in areas with a high mixture of substances, which approximates the actual situation. The results in Figs. 17 and 18 confirm the effectiveness and practicability of the proposed approach.

V. Conclusion and Future Lines

This article introduces a novel DL framework called "DAAN" that is designed to achieve unsupervised nonlinear unmixing. The proposed approach is based on the MLMM and develops a hierarchical expression to consider the multiple scattering during imaging, which can obtain more accurate unmixing results than existing methods. Moreover, we design a Gaussian constraint for the proposed DAAN to aggregate spatial correlations, achieving the integration of spatial and spectral information to enhance unmixing performance.

The framework comprises two encoders (Encoder 1 and Encoder 2) and a decoder. Encoder 1 employs a combination of 3-D convolutions and a Gaussian constraint to address the estimation of endmembers and abundances, while Encoder 2 focuses on predicting the impact of multiple scattering, with the requirement of being driven by Encoder 1 during training. The Gaussian constraint enhances the smoothness of abundances by aiding convolutions in capturing local correlations among homogeneous pixels. The decoder is employed for BSS to obtain endmembers and abundances. In the proposed DAAN, a Softmax activation function is utilized to ensure the fulfillment of abundance nonnegativity ANC and ASC. Several experimental results validate the outstanding performance of our new DAAN, regardless of whether the images are synthetic or real. The proposed DAAN framework provides a new path to use the MLMM idea and can implement a blind nonlinear unmixing framework considering multiple scattering. Our future work will continue to optimize the DL network to improve its predictive capabilities.

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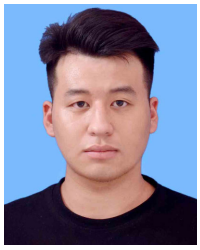
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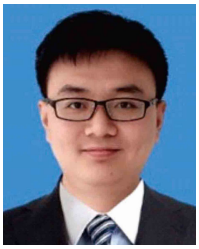
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